



Automatic Control II Project

Submitted by:

Ahmed Emad Mohamed 120220082

Islam Abdulnasser 120220034

MTE Department

To Prof. Ayman Nada

Control of a 1-DOF Helicopter Using State-Space Control with Kalman Filtering and Gravity Compensation

1. Introduction

The control of small-scale helicopters is a challenging problem due to their inherently unstable dynamics, actuator constraints, and sensor noise. In this project, a 1-degree-of-freedom (1-DOF) helicopter is controlled to maintain a desired pitch angle using a combination of state-space control, integral action, and feedforward gravity compensation. The system leverages a Kalman filter to fuse noisy accelerometer and gyroscope measurements from an MPU6500 sensor to provide accurate state estimation. The control signal is then sent to an electronic speed controller (ESC) driving a brushless DC (BLDC) motor.

This report presents the theoretical foundations, mathematical derivations, and implementation details of the control system, along with a detailed discussion of tuning, practical issues, and experimental results.

2. System Overview

2.1 Helicopter Dynamics

The 1-DOF helicopter can be modeled as a rigid body rotating around its pitch axis. The dynamics are governed by the following equation:

$$J\ddot{\theta} + b\dot{\theta} + mgl\sin(\theta) = \tau$$

Where:

- θ — pitch angle (rad)
- $\dot{\theta}$ — angular velocity (rad/s)
- $\ddot{\theta}$ — angular acceleration (rad/s²)
- J — rotational inertia of the helicopter
- b — damping coefficient of the rotor/motor
- m — mass of the helicopter
- g — gravitational acceleration
- l — distance from pivot to center of mass
- τ — control torque applied by the motor

The goal is to control θ to a desired reference θ_{ref} despite sensor noise and unmodeled disturbances.

2.2 Sensors and State Estimation

The helicopter is equipped with an MPU6500, which provides:

- Gyroscope (ω) — measures angular velocity ($\dot{\theta}$)
- Accelerometer — provides an angle estimate ($\theta_{acc} = \arctan(\frac{a_y}{a_z})$)

Both sensors are noisy:

- Gyroscope suffers from drift over time.
- Accelerometers are noisy but stable at low frequencies.

To obtain a reliable estimate of the pitch angle and angular velocity, a **Kalman filter** is used.

3. Kalman Filter Design

3.1 State-Space Formulation

We define the state vector:

$$\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

The continuous-time state-space model is:

$$y = [1 \quad 0]\mathbf{x} + v$$

Where:

- $\mathbf{w}$ — process noise
- v — measurement noise

Discretizing with a sampling time , we approximate:

$$\begin{aligned} \theta_{k+1} &= \theta_k + T_s \dot{\theta}_k \\ \dot{\theta}_{k+1} &= \dot{\theta}_k \end{aligned}$$

3.2 Kalman Filter Equations

The Kalman filter has **prediction** and **update** steps:

Prediction:

$$\begin{aligned} \hat{\theta}_{k|k-1} &= \hat{\theta}_{k-1} + T_s \dot{\theta}_{k-1} \\ \hat{\dot{\theta}}_{k|k-1} &= \hat{\dot{\theta}}_{k-1} \end{aligned}$$

Covariance update:

$$P = APA^T + Q$$

Where:

- $P = \begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix}$ — covariance matrix of the state
- $A = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}$ — discrete-time system matrix
- $Q = \begin{bmatrix} Q_{angle} & 0 \\ 0 & Q_{rate} \end{bmatrix}$ — process noise

Update:

$$\begin{aligned} y_k &= \theta_{acc} - \hat{\theta}_{k|k-1} \text{ (innovation)} \\ K &= PH^T(HPH^T + R)^{-1} \text{ (Kalman gain)} \\ \hat{\mathbf{x}}_k &= \hat{\mathbf{x}}_{k|k-1} + Ky_k \\ P &= (I - KH)P \end{aligned}$$

Where $H = [1, 0]$ and $R = R_{angle}$ is the measurement noise.

Illustration of Covariance Matrix P:

$$P = \begin{bmatrix} \text{Var}(\theta) & \text{Cov}(\theta, \dot{\theta}) \\ \text{Cov}(\dot{\theta}, \theta) & \text{Var}(\dot{\theta}) \end{bmatrix}$$

- P_{00} — uncertainty in angle
- P_{11} — uncertainty in angular velocity
- $P_{01} = P_{10}$ — correlation between angle and velocity

3.3 Ts and its Role

T_s is the **sampling period** (time between loop iterations). It is critical for:

- Discretizing continuous dynamics
- Scaling the effect of gyro integration ($\theta_{k+1} = \theta_k + T_s \dot{\theta}_k$)
- Adjusting covariance evolution over time

In our implementation:

$$T_s = \frac{1}{\text{LOOP_HZ}}$$

4. State-Space Control

4.1 Control Law

A linear state-feedback controller is applied:

$$u = -k_1 e_\theta - k_2 e_{\dot{\theta}} - k_i \int e_\theta dt + u_{ff}$$

Where:

- $e_\theta = \theta - \theta_{ref}$
- $e_{\dot{\theta}} = \dot{\theta}$
- u_{ff} — feedforward term compensating gravity
- k_1, k_2, k_i — proportional, derivative, and integral gains

4. State-Space Control

4.1 Control Law

A linear state-feedback controller is applied:

$$u = -k_1 e_\theta - k_2 e_{\dot{\theta}} - k_i \int e_\theta dt + u_{ff}$$

Where:

- $e_\theta = \theta - \theta_{ref}$
- $e_{\dot{\theta}} = \dot{\theta}$
- u_{ff} — feedforward term compensating gravity
- k_1, k_2, k_i — proportional, derivative, and integral gains

As θ increases, torque required to maintain the angle decreases: $\tau_{grav} = mgl\cos(\theta)$

Feedforward adds this torque as a base, letting the feedback controller correct only deviations

4.3 Integral Action

- $\theta_{integral} = \int e_{\theta} dt$
- Addresses **steady-state error** caused by:
 - Motor nonlinearity
 - Gravity
 - Sensor bias

Integral gain k_i is tuned small to prevent overshooting and oscillations.

4.4 Conversion to ESC PWM

- Control output u is a **torque deviation**
- Converted to PWM signal:

$$PWM = PWM_{hover} + u \cdot gain$$

- PWM_{hover} — experimentally found value where helicopter hovers without tilting
- $gain$ — scaling factor (18 in code) to map control to microseconds

5. Implementation Notes

5.1 Sensor Offset

At rest, accelerometer may not read zero due to mounting or bias. This is corrected:

$$\theta_{corrected} = \theta - \theta_{offset}$$

- Offset is read once at startup
- Corrects the reference frame for proper control

. Controller Concept

- **Proportional (P)** — reacts to error in angle
- **Derivative (D)** — reacts to angular velocity (damping)

- **Integral (I)** — eliminates steady-state error
- **Feedforward** — compensates constant gravity torque

Combined, the **PID + feedforward** ensures:

- Stable approach to desired angle
- Minimal overshoot
- Correct steady-state angle despite bias or gravity

8. Matrix Illustration

State Vector:

$$\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

System Matrix (discrete A):

$$A = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}$$

Covariance Matrix:

$$P = \begin{bmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{bmatrix} = \begin{bmatrix} \text{Var}(\theta) & \text{Cov}(\theta, \dot{\theta}) \\ \text{Cov}(\dot{\theta}, \theta) & \text{Var}(\dot{\theta}) \end{bmatrix}$$

Noise Matrices:

$$Q = \begin{bmatrix} Q_{angle} & 0 \\ 0 & Q_{rate} \end{bmatrix}, R = R_{angle}$$

9. Tuning Guidelines

- k_1 — proportional: start small, increase for faster response
- k_2 — derivative: adjust for damping, prevent overshoot
- k_i — integral: start very small (0.05–0.7) to correct steady-state error
- Q, R — Kalman noise: increase Q_{angle} for trusting gyro more, increase R for trusting accelerometer less

10. Conclusion

The 1-DOF helicopter control system successfully integrates:

- **Kalman filtering** to fuse accelerometer and gyroscope data
- **State-space PID control** for stable angle tracking
- **Gravity feedforward** to remove steady-state bias
- **ESC PWM mapping** to apply motor torque

This approach provides accurate, smooth control of pitch angles in the presence of sensor noise and model uncertainties. The implemented equations, matrices, and control laws are fully consistent with control theory and practical small-scale helicopter dynamics.